

Builder Basics — The Living Village (Extension · Advanced)

Topic: Bring It All Together · **Course:** Builder Basics · **Level:** Advanced · **Time:** about 90 minutes

🧩 Bonus challenge, after **Complete Your Village**. Here you use **loops**, **variables** for sizes, and **one function you call more than once**.

1 · The Build 🎯

Two houses already stand: the **farm house** 🏡 and a tall house 🏠. Add a **watchtower** on the tall house, then build the other six parts with code.

Every part must meet its spec:

Part	Spec
🗼 Watchtower	on the tall house, ≥ 4 blocks taller than it
🌾 Farm	2 crop types , planted in rows by a loop
🧱 Fence	fully encloses the farm or the animals — no gaps
💧 Pond	a dug dip filled with water — measure its width W
🐟 Fish	≥ 3 fish in the pond
🌉 Bridge	spans the pond: length = W
🕒 Well	square rim, water inside
🐄 Animals	≥ 4 , inside the fence

⚠️ **Rules:** loops for anything repeated · **variables** for sizes · **one function** you call for **2 or more** parts · only animals are spawned, everything else is code.

2 · Plan the Map 🗺️

This is your village from above. 🏠 = farm house, 🏡 = tall house. **x** goes across, **z** goes forward.

	1	2	3	4	5	6
6						
5		🏡				
4						
3					🏠	
2						
1						

Draw the other six parts on the map. Write the (x, z) corner where each one starts.

Bridge length = pond width W. What is your W, and how did you measure it?

3 · Loops, Variables, Functions

The one function you call twice or more — what does it build, and what does it take in?

Which sizes are variables? Name two and why.

4 · Show Your Work 📷 🎥

Build all eight parts to spec, then move the agent between them so nothing overlaps.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.